

If Else Questions

1. Largest Among Three Numbers

Write a C program to input 3 numbers and print the largest number using `if-else`.

2. Leap Year Checker

Write a program to check whether a given year is a leap year or not.

📌 Rules:

- Divisible by 4
 - But not divisible by 100
 - Unless divisible by 400
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3. Electricity Bill Generator

Create a program to calculate electricity bill based on units:

Units	Rate
0–100	₹5/unit
101–200	₹7/unit
201+	₹10/unit

Add 18% GST at the end.

4. Grade System

Input marks of 5 subjects and calculate percentage.

Print:

- A Grade → 90+
- B Grade → 75+
- C Grade → 50+
- Fail → below 50

Also print:

- “Distinction”
 - “Pass”
 - “Fail”
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5. ATM Withdrawal System

Input:

- Account balance
- Withdrawal amount

Conditions:

- Minimum balance should remain ₹1000
 - Withdrawal should be multiple of 100
 - Print success/failure message
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6. Login Authentication System

Create a simple login system.

Correct credentials:

- Username: `admin`
- Password: `1234`

Use nested `if-else`.

7. Triangle Validator

Input 3 angles.

Check:

- Whether triangle is valid
 - Then print:
 - Equilateral
 - Isosceles
 - Scalene
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8. Employee Salary Bonus System

Input:

- Salary
- Years of experience
- Performance rating

Conditions:

- Experience > 5 years → 20% bonus
- Rating A → Extra 10%
- Rating B → Extra 5%
- Otherwise no extra bonus

Print final salary.

9. E-Commerce Discount System

Create a shopping bill program.

Conditions:

- Above ₹5000 → 20% discount
- Above ₹2000 → 10% discount
- VIP customer → extra 5%

Add GST after discount.

10. Online Voting System

Input:

- Age
- Citizenship (**Indian** or not)
- Criminal record (**Yes/No**)

Eligible only if:

- Age ≥ 18
 - Indian citizen
 - No criminal record
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11. Smart Traffic Signal System

Input vehicle speed.

Conditions:

- Speed < 40 → Safe
 - 40–80 → Normal
 - 81–120 → Warning
 - Above 120 → Challan + License Suspension
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12. Income Tax Calculator

Calculate tax based on income slabs:

Income	Tax
Up to ₹2.5L	0%
₹2.5L–₹5L	5%
₹5L–₹10L	20%
Above ₹10L	30%

Add 4% cess.

13. Railway Ticket Booking

Input:

- Age
- Gender
- Ticket class

Conditions:

- Senior citizen discount
- Female discount
- AC class extra charge

Print final fare.

14. Student Result Management System

Input marks of:

- C
- Java
- DBMS
- HTML
- React

Conditions:

- Any subject below 35 → Fail
- Otherwise calculate:
 - Percentage
 - Grade
 - Topper status (above 90%)

Use nested conditions.

15. Bank Loan Eligibility System

Input:

- Salary
- CIBIL score
- Existing loan
- Age

Conditions:

- Salary > ₹25,000
- CIBIL > 750
- Existing loan < ₹5L
- Age between 21–60

Print:

- Eligible
- Not Eligible
- Premium Customer Loan Offer

For Loop Questions

1. Factorial of Number

Calculate factorial using `for` loop.

Example:

$$5! = 120$$

2. Reverse Table

Print multiplication table in reverse order.

Example:

$$5 \times 10 = 50$$

$$5 \times 9 = 45$$

3. Count Digits

Input a number and count total digits using loop.

4. Sum of Digits

Find sum of digits of a number.

Example:

$$1234 \rightarrow 10$$

5. Reverse Number

Reverse a number using loop.

Example:

$$1234 \rightarrow 4321$$

6. Strong Number

Check whether number is Strong Number.

Example:

$$145 \rightarrow \text{Strong Number}$$

7. Find HCF and LCM

Find HCF and LCM of two numbers using loops.

8.Perfect Number

Check whether a number is perfect or not.

Example:

$$6 = 1 + 2 + 3$$

9.Decimal to Binary Conversion

Convert decimal number to binary using loop.

10.Menu Driven Calculator

Create calculator using:

- `for`
- `switch`
- loops

Features:

- Addition
- Subtraction
- Multiplication
- Division
- Continue option

11.Student Result System

Input marks of 5 students using loop and print:

- Total
- Percentage
- Grade

12.Employee Salary System

Input salary of multiple employees and calculate:

- Bonus
- Tax
- Final Salary

Using loops + conditions.

13. Online Exam Result System

Input marks of 5 subjects for multiple students.

Using loop:

- **Calculate:**
 - **Total**
 - **Percentage**
 - **Grade**
- **Print Pass/Fail**

14. Mobile Recharge System

A shopkeeper recharges multiple customers.

Recharge Plans:

- ₹199
- ₹399
- ₹599

Using loop:

- Input customer recharge
- Calculate total earnings of shopkeeper

15. Hospital Patient Temperature Monitor

A hospital records temperature of patients.

Using loop:

- Input temperature of 20 patients
- Count:
 - Normal patients
 - Fever patients ($>100^{\circ}\text{F}$)